

# Preprocessing and Initial Model Report

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## Basics

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Title: Will Bridging The Digital Divide Bring Prosperity To The Second “Next Billion Users” And Their Home Countries?

GitHub: <https://github.com/Merrimack-Capstone/MSDS-capstone>

## Recap: Background & Question

We are posing the following research question: To what extent has growth in internet access of populations influenced population wellbeing in more developed nations, and can that be used to predict improvements in wellbeing and economic progress among global minority populations (the "next billion users")?

**Our hypothesis is as follows:** Increased access to the internet in the developing world is a measurably material contributing factor to improved population wellbeing - so much so that we may be able to predict how much wellbeing will improve based on increased digital inclusion.

**Our predictions:** We expect that digital inclusion will be moderately-to-strongly correlated with overall wellbeing. We expect that PREDICTING the rate of improvement in wellbeing based on

internet inclusion alone will be extremely challenging due to the number of other factors influencing wellbeing, but hope to be able to isolate its impact on wellbeing. We expect that clustering may help lead to better predictive performance, if we have time to get to it.

Methods

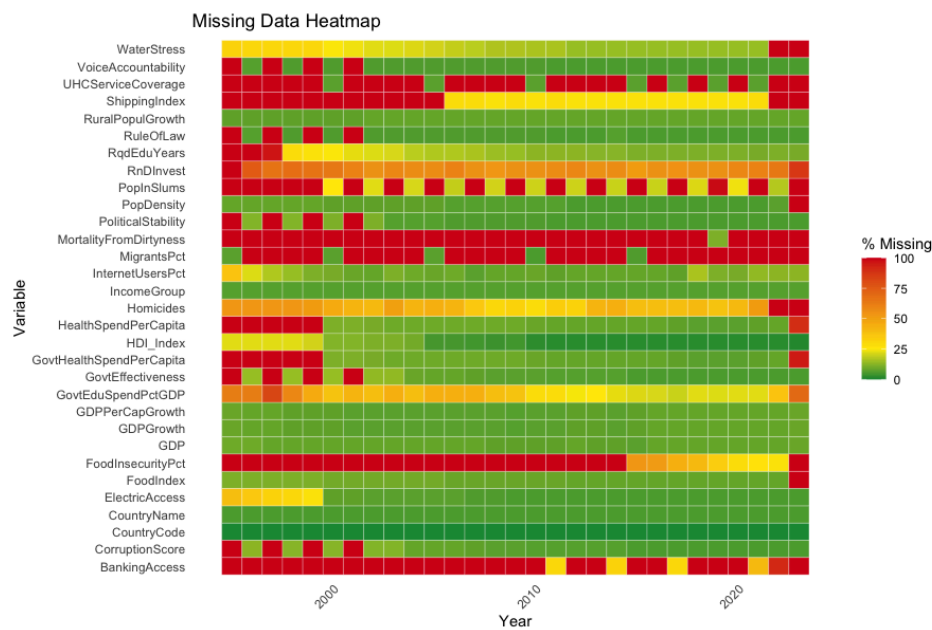
Approach and Rationale

Feature Creation

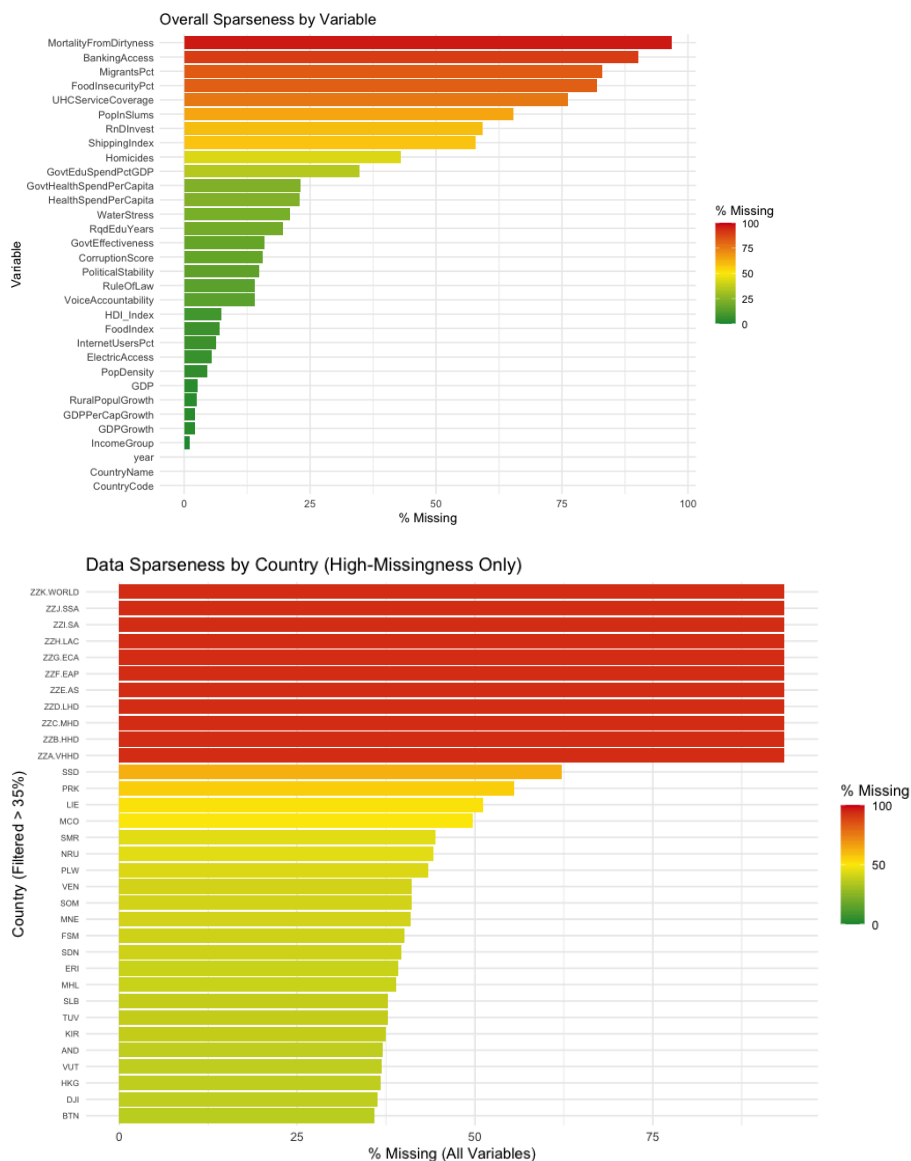
We created two new features that represent the year-on-year change in internet access and HDI, through simple calculations that take advantage of the lag() function in R in order to use changes rather than absolute levels in our models. We also pulled in internet access data from 1 and 2 years prior as additional features, and calculated a 3 year cumulative change internet access feature, since our view is it could take up to 3 years for changes in digital inclusion to translate into improvements in HDI.

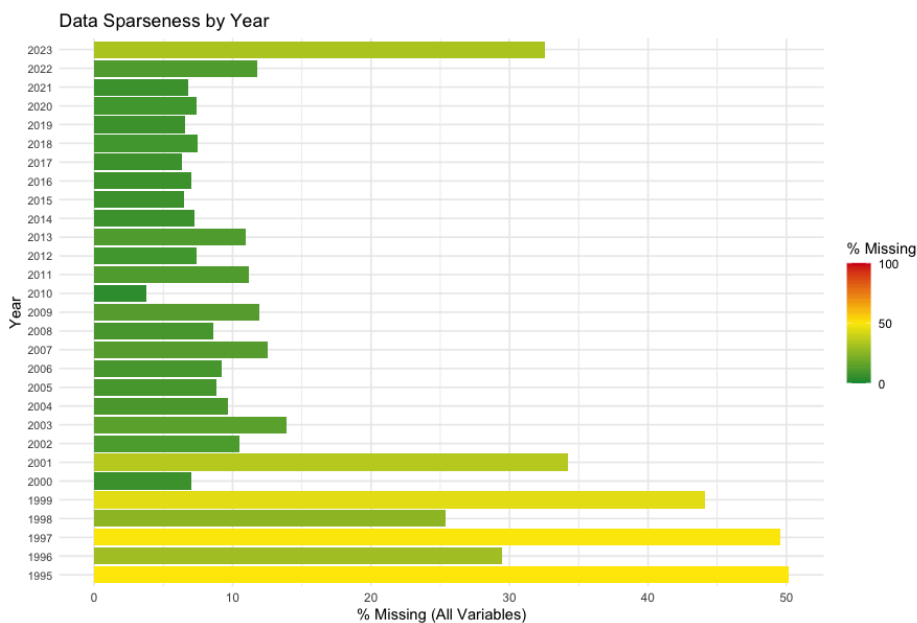
Data Dropping

We began by first creating a heatmap of data “missingness” across variables and years, which was used iteratively throughout the data dropping and imputation. The starting point for data missingness, which was quite material, is below:



To determine what data to drop, we applied an overarching philosophy that individual variables, years, and countries with more than 40% of their data missing should automatically be dropped. We made extensive use of visualizations to drive this process, examples of which are below - the visualizations guided the immediate dropping of quite a few “slices” of data as being too sparse to be salvageable via imputation.





### ***Training / Test Data Split***

After dropping these data points / observations, we considered the question of when and how to split the data into training and test sets using a 70/30 ratio. This is an important question when contemplating data imputation, handling of skewness, or PCA analysis, since we would not want data leakage to occur between the test and training data sets. When preparing a dataset for a machine learning model, it's imperative for the reliability of the outcome of the models to split the data into training and testing sets before performing imputation to prevent data leakage. Data leakage occurs when information from the test set "leaks" into the training process which can lead to an overly optimistic evaluation of the model's performance. If missing values are imputed, data is normalized, or principal component "loadings" are created using data from the entire dataset, or if data in the test set is correlated to data in the training set, then the training set will be influenced by the information present in the test set. By splitting the data first, you ensure that no such leakage occurs, which ensures a less biased and more reliable evaluation of the model's generalization ability on unseen data.

Time series panel data such as ours presents particular issues when splitting because much of the data is “autocorrelated”, meaning that a particular observation is likely to be highly correlated to observations immediately preceding or following it. In our case, our time series data is by country and year, so, as an example, internet access percent in 2020 for Angola is likely to be closely related to internet access for Angola in 2019 and 2021. This means that if the data is split at random and a particular country’s data, or a particular year’s data, is split across the training and test data sets, this would create leakage issues. We considered two methods for splitting the data:

- 1) Keeping entire year’s data and so splitting at a point in time (e.g. keeping 2000-2019 in training and 2020-2023 in test)
- 2) Keeping entire country’s data and so splitting at the country level (e.g. keeping 150 or so countries’ complete time series data in training and the remaining 50 or so in test)

Option 1 is more appropriate when you are actually FORECASTING in your predictive modeling exercise, using time (in our case, year) as one of your predictors. This is not our method - we are simply trying to predict any given country’s HDI based on internet access (as well as other factors in a multi-variate model) at a single point in time. So we chose option 2, electing to keep entire countries’ data together, and split the countries up 70/30 with 70% of the countries appearing in the training data and the remaining 30% in the training data. To avoid bias, we stratified the countries by IncomeGroup so that the same rough distribution by IncomeGroup was maintained in both data sets - this was done based on the EDA analysis that showed distinct differences in HDI and internet access based on Income Group.

This method of splitting the data then dictated which components of our remaining feature engineering needed to be done pre-split versus post-split for leakage avoidance, as follows:

- *Data Imputation*: PRE-SPLIT - Since all of our imputation is done only inter-country (as described below) using other time periods for the same country, and each country’s entire data

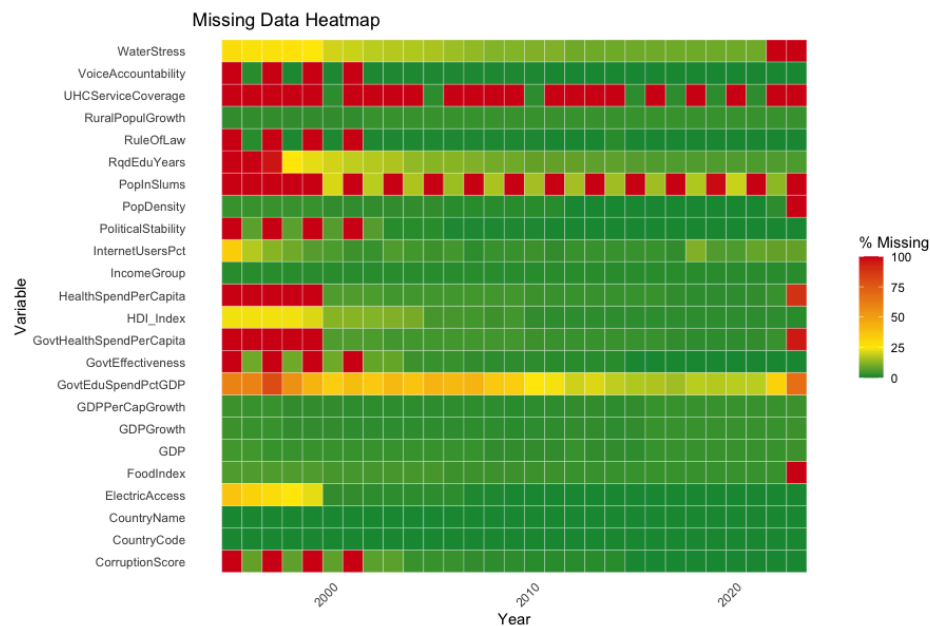
set is self-contained within either training or test data, there is no risk of leakage by doing intra-country data imputation on the entire data set pre-split.

- *Skewness*: POST-SPLIT - We used Yeo-Johnson normalization (see below) to handle the severe negative skew of some of our data. This needs to be done separately on training versus test data to avoid leakage - the lambdas derived from the training data are used to transform both the training and the test data
- *PCA*: POST-SPLIT - This also needs to be done separately on training versus test data to avoid leakage - the principal component loadings derived from the training data are used to create Principal Components for both the training and test data

### ***Data Imputation for missing values in the entire data set***

We first tried to use K-Nearest Neighbors (KNN) imputation, which is a powerful method for handling missing data that estimates missing values based on the values of its "k" nearest neighbors. For each missing value, the algorithm identifies the k most similar data points in the dataset based on a distance metric like Euclidean distance and then imputes the missing value using the weighted average of the values from those neighbors. This method was attempted using k=3 and k=5, but the attempts were unsuccessful as a means of imputation due to high processing times which impacts replicability and also due to the output not producing the intended outcome. This approach is more sophisticated than simple methods like mean or median imputation, but a significant drawback of KNN imputation is its high computational cost because the algorithm must calculate the distance between missing values across all missing data in the full set (Kaur, 2025).

We then looked for patterns in the missing data and were pleased to see those patterns emerge in the heatmap below with the strategy used, as follows:



- We used linear interpolation to impute in the missing years in Voice Accountability, RuleOfLaw, PopInSlums, Political Stability, GovtEffectiveness and CorruptionScore, and the later years of UHCServiceCoverage, where data was only reported every other year - a perfectly justifiable approach given the short time period spanning each interpolation
- We also used linear interpolation to fill in the larger data gaps in earlier years between 2000 and 2015, where data was reported only every five years, but only after demonstrating that the trajectory of the data over those twenty years was a virtually straight line, justifying that approach.
- We used linear interpolation to fill in the missing 2001 “governance” variables which were oddly not reported - again justifiable given the short period spanned

After filling in patterned missing data, we imputed the rest using the following approach:

- We extrapolated the missing data for 2023 using quadratic extrapolation of all the prior country/year pairs - a notable flagged assumption worth watching

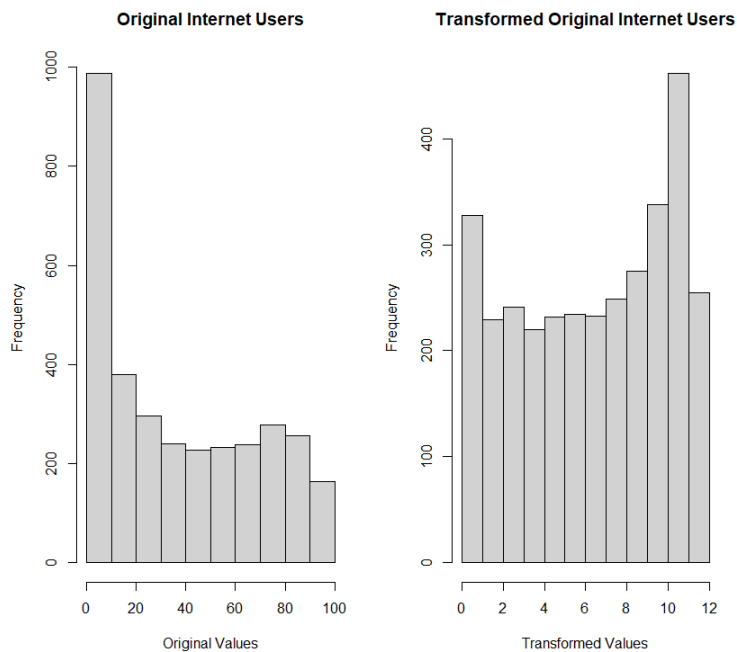
- All other data was imputed using the following method, applied using data strictly from the same country for other years to avoid cross-country data contamination:
  - If 3 or more yearly data points were available for the variable, we applied quadratic regression-based imputation
  - If 2 yearly data points were available, we applied linear regression-based imputation
  - If only 1 data point was available, we replicated that value across all yearly observations for that country.
  - If no other data was available it was left NA
- It's important to note that while this approach is less defensible, the missingness it was applied to was quite sporadic, and without it, dropping rows with NA values before modeling would have cut our data set more than in half.
- This is true except for GovtEduSpendPctGDP which had more substantial missing data (25%), but that we felt merited salvaging given that education is one of three components of our HDI response variable and this is one of only two education-related predictors we have.
- Regardless, we will monitor the effect of this estimation and run models with and without the estimated data as time permits

### *Skewness*

Yeo-Johnson was chosen as a method to address skewness in the internet usage features (see results on training data below) that were determined to need improvement when it came to normal distribution for better accuracy when running the predictive and machine learning models during the upcoming phase of this analysis. This method works by inflating lower variance and deflating higher variance to create a more evenly distributed set of data (Ultrahaus, 2023). It was chosen over Box-Cox even though no negative numbers were needed to work with in the event that negative numbers were present and



overlooked. Yeo-Johnson as a method is very time efficient and does not take long to process which is an important factor to consider when taking reproducibility into account.

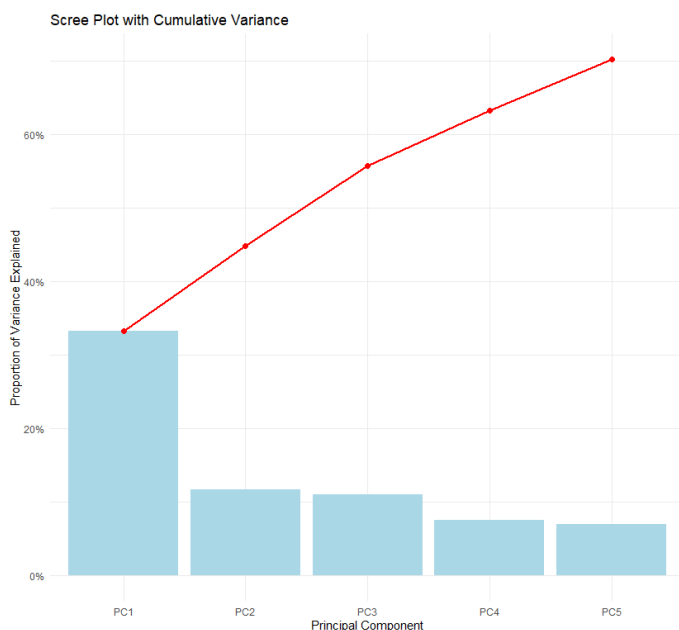


## ***Binning***

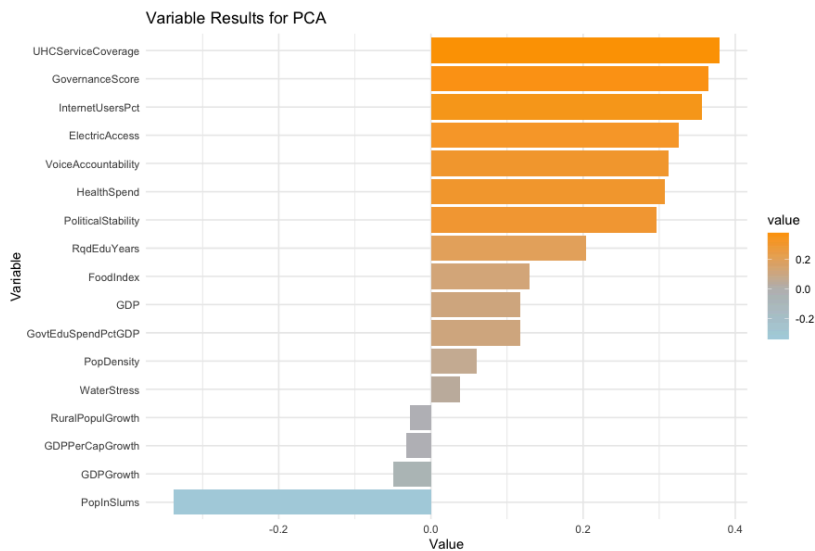
The creation of new fields in our dataset was an important part of feature engineering. By binning features into a calculated column, our team was able to address some of the factors that were showing as 90% correlated. This was done AFTER imputation to ensure that missing data was handled before bins were created, thereby avoiding skewing bins and/or excessive NA binned values. The creation of two columns into calculated average scores of a governance accountability indicator rating and a field that calculates spending on public health initiatives allows for these features to be included in future models without impacting the results as greatly due to the extremely high correlation amongst the original features. A binned HDI Category was also created as a column during feature engineering to produce the field of categories that the UNDP uses as ranking of HDI for various nations. These categories encompass a range of scores that translates to “Low human development,” “Medium human development,” “High human development,” and “Very High human development.”

## *Unsupervised learning*

PCA was chosen as the unsupervised model because it handles multicollinearity well and can be used for dimensionality reduction when there are a large set of features to consider. Principal Component Analysis (PCA) is a reliable unsupervised dimensionality reduction technique used to reduce the number of features in a dataset while retaining as much information as possible. It works by transforming the data into a new set of dimensions, called principal components, which are uncorrelated to each other. The components in PC1 and PC2 have the highest variance contribution, with variance decreasing in subsequent components, which helps produce more consistent results in future model training and predictions (Geeks for Geeks, 2025b). This process can help to visualize high-dimensional data, reduce noise, and speed up the training of machine learning models by simplifying the dataset without losing significant information. The loadings created by PCA are the coefficients or weights that explain how the original variables contribute to each principal component. The principal components, which can be used with tree-based models such as Random Forest, are now the new, transformed features that serve as the input data, with less dimensionality for machine learning model (Geeks for Geeks, 2025a). As can be seen in the chart below, 70% of the variance across our predictors can be explained using the top 5 PC's, which would then reduce model dimensionality.



An even more important finding from the PCA exercise is that the first Principal Component (PC1) shows a 92.5% correlation with our HDI response variable - and, as the below exhibit shows, Internet Usage is the third most heavily weighted factor in PC1, which bodes well for our models proving out our hypothesis regarding the connection between internet access and HDI.



Finally, The PCA “biplot” to the left shows that Principal Component 1 (PC1) (within which internet access is a key feature) is a dominant factor in determining which HDI category a country falls into - further solidifying that our hypothesis is likely

## Preliminary Model Methods and Evaluation Process

### Part 1: Two-variable models

As we laid out in our modeling plan, we first built a few two-variable models using internet access (as predictor) and HDI (as response) to determine if we can accurately predict changes in well-being using solely changes in internet access. The first model we ran was a simple linear regression using InternetUsersPct as our independent variable and HDI\_Index as our dependent variable (these are annual “point-in-time” measures”. Recall that during our EDA exercise we showed that there was a correlation between the two, as shown below, with an  $R^2$  of 0.524 - this was our baseline. Our model fared better:

| r.squared | adj.r.squared | sigma  | statistic | p.value | df    | logLik | AIC    | BIC    | deviance |
|-----------|---------------|--------|-----------|---------|-------|--------|--------|--------|----------|
| <dbl>     | <dbl>         | <dbl>  | <dbl>     | <dbl>   | <dbl> | <dbl>  | <dbl>  | <dbl>  | <dbl>    |
| 0.661     | 0.660         | 0.0932 | 5524.     | 0       | 1     | 2712.  | -5419. | -5401. | 24.6     |

The trained model had an  $R^2$  of 0.661, meaning that 66.1% of the variance in HDI can be explained through InternetUserPct. The improvement in the model versus our EDA results stems from having normalized the internet access predictor to reduce the severe left-skew with data clustered at or near 0%. Also encouraging is that the model generalized well to data it had not seen, with an  $R^2$  of 0.634 (see right) and so very

| .metric | .estimator | .estimate |
|---------|------------|-----------|
| <chr>   | <chr>      | <dbl>     |
| 1 rmse  | standard   | 0.0955    |
| 2 mae   | standard   | 0.0770    |
| 3 rsq   | standard   | 0.634     |

similar results versus the test data - this model, as simple as it is, supports our general hypothesis, since “A R-squared between 0.50 to 0.99 is acceptable in social science research especially when most of the explanatory variables are statistically significant.” (Ozili 2023). HOWEVER - as we suspected, this model alone is not sufficient to predict HDI - since HDI is on a scale of 0 to 1, an RMSE of 0.0955 and MAE of 0.077 are both material average prediction errors - more than sufficient an average “miss” to place the predicted HDI entirely in the wrong UNDB-defined category (seen in our data as HDI\_Category with values of Low, Medium, High and Very High). So, while we have demonstrated ample correlation between internet access and HDI, the predictive value of this model is mediocre. As can be seen in the below scatterplot, errors are fairly substantial across the range of HDI’s observed. In addition, the model consistently overestimates HDI in underdeveloped countries and underestimates it in well developed countries. This could potentially be addressed by producing separate or piecewise regressions based on Income Group or other breakpoints, by using polynomial or splined regression, or other means. This was tabled to later.

### **Model Specifics:**

Model type: OLS ( $\text{HDI\_Index} \sim \text{InternetUsersPct}$ )

Assumptions: Approx. linear relationship after transform; residuals with roughly constant variance; no single point dominates; country-level independence across splits as demonstrated.

Checks: Scatter + 45° line; residuals-vs-fitted;

Takeaway: Moderate fit; systematic under- at high HDI and over- at low HDI suggests mild nonlinearity/omitted variables.

Since our linear regression model’s RMSE was large enough to limit predictive power, we ran another simple logistic regression, using InternetUsersPct as our independent variable and HDI\_Category as our dependent variable - similar to the first model, but now attempting predicting the broad category (low,

medium, medium high or high) of HDI based on internet access. This model also showed the influence that internet access has on HDI:

- The model's predictive accuracy using the test data was 56.4%; While that may not seem very good, it is still better than totally random - with four equally likely categories, random guesses would have a 25% accuracy rate so this is a big improvement.
- The distribution of the data is NOT equal - the biggest class is Very High at 28.3% - so guessing entirely the dominant class would yield a 28% accuracy rate, which our model doubles

| High     | Low      | Medium   | Very high |
|----------|----------|----------|-----------|
| 24.87741 | 23.47172 | 23.30827 | 28.34260  |

However - while this model also proves our basic hypothesis around the correlation between internet access and HDI, its predictive value is also poor. Not only is accuracy relatively low at 56%, but if we look at the confusion matrix (right), we see that errors can be extreme, by missing by more than one level (e.g. predicting High or Very High when the actual HDI level was Low). This could potentially be improved by tweaking the model, or using a different two-variable modeling method altogether - which was tabled to later.

| Actual    | Predicted |        |      |           |
|-----------|-----------|--------|------|-----------|
|           | Low       | Medium | High | Very High |
| Low       | 201       | 67     | 22   | 3         |
| Medium    | 97        | 84     | 71   | 15        |
| High      | 42        | 53     | 105  | 119       |
| Very High | 0         | 14     | 63   | 343       |

We ran several additional two-variable models, to see if using CHANGE in internet access vs CHANGE in HDI, or if using a lagged internet percentage, or using a quadratic model, would produce better results. However, we found that the results either did not improve (quadratic model) or were nonsensical (the others) because we would need to do some additional feature engineering for these models to function properly. This was also tabled.

### Model Specifics:

Model Type: Multinomial Logistic (HDI\_Category ~ InternetUsersPct)

Assumptions: Linearity; no perfect separation; reasonable per-class support, country-level independence across splits as demonstrated, multinomial (not ordinal) response

Checks: Accuracy and confusion matrix (note errors are mostly adjacent-class).

## Part 2: Multi-variate machine learning models

Given that the limitations of the two-variable approach we expected did in fact materialize, we proceeded with the next phase of our modeling plan, which was to build multi-variate predictive models that could better predict HDI, and then isolating the importance of internet access within those models. While we attempted to reduce collinearity in our data through the use of binning, we remain concerned that much of our predictors are still collinear enough to make statistical modeling challenging. As a result, we focused our multi-variate modeling efforts on machine learning models, which tend to ignore collinearity and de-emphasize unimportant features. We first attempted to produce reliable results through SVM as a predictive model on our data, but it was ultimately determined that due to the high inaccuracy and long processing time of the model, even after adjusting parameters, other models would be better suited for our analysis and that it would not be included in the final code. XG-Boost and Random Forest were then evaluated, with better results. (It is imperative that a variety of models are tested before coming to conclusions about the performance metrics of other models.) We elected to focus on XGboost given more processing preliminary outcomes.

To improve on the two-variable logistic regression classification model we produced, we built and optimized an XGBoost classification model that uses all of the predictors (except Principal Components) to predict HDI\_Category. Iterative loops were implemented to systematically build and evaluate a series of models using various values for the eta, maximum depth and subsample hyperparameters. We did not vary the number of rounds - instead we chose a large number of rounds (3000) and specified an early-stop parameter (if mean error has not improved) of 10 rounds. For this classification, a multi-class

loss function was utilized to predict probabilities of class membership (Brown Lee, 2021). To minimize overfitting of the model, cross-validation via the k-fold method was used to find the optimal parameters for the predictive model to pass to the final training model (Geeks for Geeks, 2025c). This dual method approach of automated hyperparameter tuning and feature selection within a single script allowed us to generate a collection of models with varying hyperparameter settings and feature subsets, ensuring that we arrived at the optimal version of this model possible with the least amount of overfitting for this phase of the analysis (Geeks for Geeks, 2025f). The resulting model was then trained on the full training dataset and evaluated on the test data.

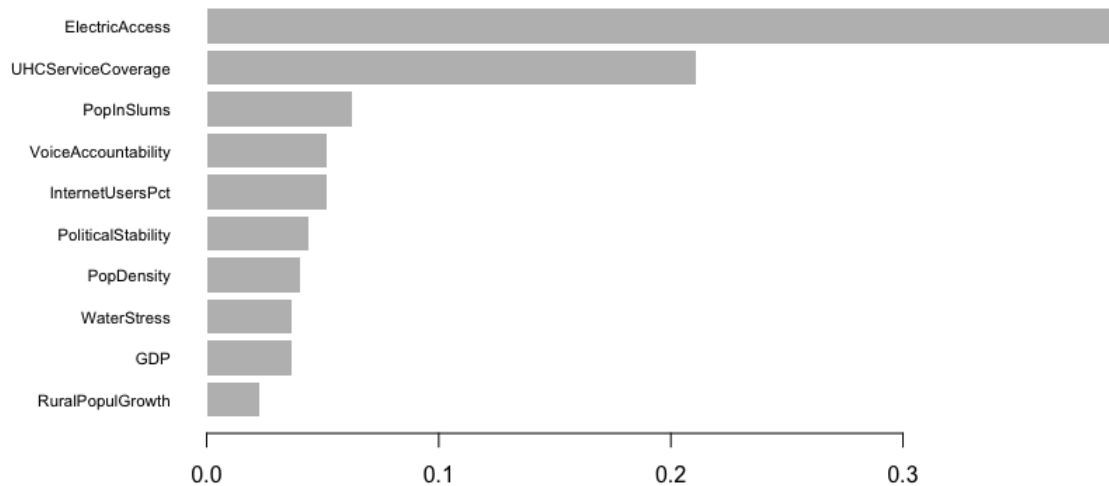
The method used to produce the folds for the k-folds cross validation merits discussion: We used five different folds in the exercise and averaged the errors across the different model/fold combinations to select the best model that also minimized overfit. However, these folds were not created completely randomly - we needed to eliminate temporal data “leakage” in the creation of the validation withheld data folds, just as we did with the training/test split. We did so the same way, by creating folds that randomly held out a portion of the data but kept entire countries’ data together (either in the training or evaluation fold).

This model proved far more accurate and useful for predicting HDI:

- Accuracy at predicting the correct HDI was 76.7% - a material improvement
- Equally importantly, as the confusion matrix to the right demonstrates, there were NO multi-level misses where the prediction was off by more than one level (e.g. Low predicted as Medium High or High). All errors placed the prediction in a level adjacent to the actual HDI level for that observation.

|        |     | Predicted |     |     |  |
|--------|-----|-----------|-----|-----|--|
| Actual | 0   | 1         | 2   | 3   |  |
| 0      | 203 | 67        | 0   | 0   |  |
| 1      | 33  | 119       | 57  | 0   |  |
| 2      | 0   | 16        | 198 | 11  |  |
| 3      | 0   | 0         | 73  | 326 |  |

The model also indirectly supports our thesis regarding the influence of internet access on HDI - a review of the Feature Importance Matrix (below) shows Internet Usage as the fifth most important feature. We can also make several other observations looking at the feature importance:



- Access to electricity is by far the most important predictor - it makes sense that this would be more important than (albeit potentially collinear with) internet access to people's wellbeing., since electricity is likely correlated to income, health AND education, the three components of HDI
- Health care coverage is far and away the second most important component - unsurprising given that health / longevity is a key component of HDI
- Two of our "political environment" variables are in the top 6 (VoiceAccountability as #4, Political Stability as #6) - showing the sound rationale of including these indicators.
- Population in Slums also ranks very highly - likely again as a proxy for low income and unhealthy living conditions

Further evaluating these results proved difficult since traditional AUC / ROC analysis is not appropriate because we have a multi-level (non binary) response variable. Further analysis was tabled pending further research into the use of alternative methods such as One-vs-Rest (OvR) or Hand-Till AUC.



## Model Specifics:

Model Type: XGBoost Classification (HDI\_Category)

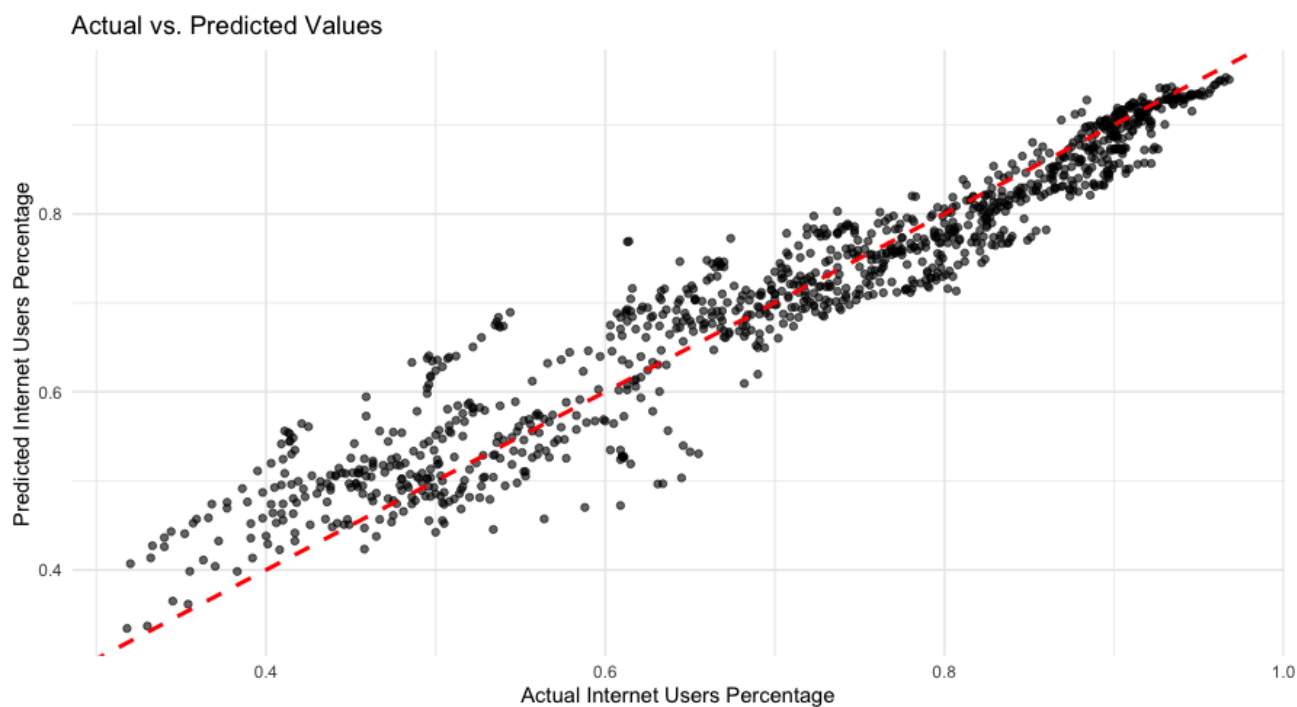
Assumptions: No major class imbalance, categorical variables factored properly, data independence via country split, sufficient trees will split the classes

Checks: Test accuracy + confusion; observed only adjacent-class errors.

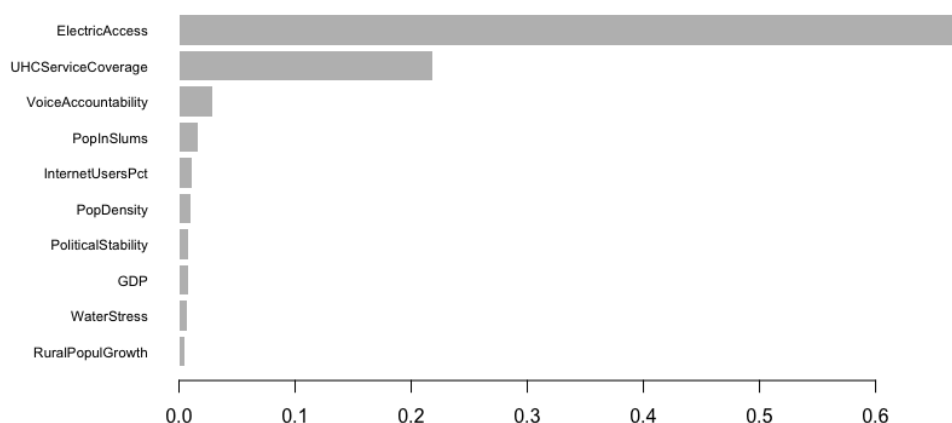
For our final preliminary model, we repeated the XGBoost exercise, but as a regression task to predict the numeric HDI\_Index using the same predictors. The goal of this model was to predict HDI\_Index while minimizing a function that measures the difference between the predicted and actual continuous values, calculated as the mean squared error. The core process of boosting remains the same as the classification version of the model, where the model iteratively builds a sequence of regression trees, where each new tree is trained to correct the residuals of the previous ones. The final prediction is the sum of the predictions from all the individual trees (Geeks for Geeks, 2025b). The same iterative process through many sets of hyperparameters was used, and the same k-folds cross validation was performed, using the same folds developed for the XGBoost classification model.

The results of this modeling exercise, particularly as it pertains to our thesis, are more complicated. On the one hand, the model does a FAR better job at predicting HDI\_Index than the two-variable model:

- The mean absolute error when evaluating the model on test data it had not seen was 0.035 which is less than half the MAE of the linear two-variable model, and small enough that prediction errors would seem to be far less likely to put the predicted HDI into a different [low, medium, medium high, high) category
- As the below scatter plot shows, predictions fall much closer to the 1x2 slope line, and while the overestimation in low development countries and underestimation in high development countries persists, it is markedly less pronounced.



On the other hand, it's unclear whether this model supports our hypothesis very well. Examining the feature importance plot (below) shows that while InternetUsersPct is still the fifth most important feature, its importance appears negligible since ElectricAccess and UHCServiceCoverage absolutely dominate the model.



### Model Specifics:

Model Type: XGBoost Regression (HDI\_Index)

Assumptions: Same as for XGBoost Classification but sufficient trees should minimize MSE

Checks: Test MAE/RMSE; predicted-vs-actual with 45° line

We ultimately chose to table the use of Random Forest and Principal Components until we can do further research. PCA was initially applied to our original features to reduce the dimensionality and create a new set of uncorrelated components that capture the maximum variance in the data. By using the most significant of these principal components as features for a Random Forest model, we may be able to mitigate multicollinearity, reduce training time, and improve predictive accuracy by de-noising the data. Without using this method, our data may not fit the requirements to successfully execute a Random Forest model due to the high dimensionality of our data. We believe that the use of a Random Forest model trained on the results of Principal Component Analysis (PCA) is a valuable technique for handling high-dimensional data. HOWEVER - we tabled this because while we expect that this model might yield greater accuracy at predicting HDI, we are unsure as to how to extract and explain the impact of internet access on the model, since it is buried within PC1. Since our thesis is specifically tied to explaining the role that internet access plays in HDI, we need to either redraw our PCA to exclude internet access, or find a way to extract the contribution of internet access to the model results.

## ***Discussion***

Most of our discussion points, rationale, and key takeaways regarding the pre-processing and feature selection process, our preliminary models, and how what we have done so far supports or refutes our hypothesis or impacts / changes our proposal and modeling plan have been laid out above and we won't repeat them here. Our results so far have been very much in line with what we expected and our modeling plan has proceeded as we have laid it out - in fact we are more optimistic about how deeply we will get into our modeling plan than our final proposal suggested. Below we have updated our modeling plan, reflecting changes in probabilities and plans.

| Phase   | Description                                            | Original Probability | Current Probability | Comments                                                                                                                                                                                                                                                                                                        |
|---------|--------------------------------------------------------|----------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 | Two-variable models (Internet vs HDI)                  | High                 | 100%                | Done (though can be tweaked and alternative models done)                                                                                                                                                                                                                                                        |
| Phase 2 | Multi-variate models, isolating importance of Internet | High                 | 100%                | First pass done - more tweaking and alternative models will be built                                                                                                                                                                                                                                            |
| Phase 3 | PCA and Clustering                                     | Medium-High          | High                | PCA has been completed. Qualitative clustering by Income Group has been done. Likely no need for another quantitative version, though clustering by electricity access may be useful                                                                                                                            |
| Phase 4 | Using PCA / Separate Models by Segment                 | Low                  | Medium              | We will make every effort to do this - running a model with PCA is straightforward though unclear hope it will support our thesis. Running different models for different segments would be extremely additive, both to reduce the under/over estimation pattern or to address the impact of electricity access |
| Phase 5 | Assess impact of fixed vs mobile internet              | Very Low             | Zero                | We don't think we have good enough data separating broadband from mobile, and we'd need to redo some feature engineering before doing this. Worthwhile goals but inadequate data and time                                                                                                                       |

There are two important observations we wish to highlight that have come about from our modeling efforts, which are both notable on their own and will change our approach to additional models (if we have time):

- 1) We did not anticipate the importance of access to electricity as a predictor of HDI. It completely dominates the XGBoost model results, and we suspect that including it in statistical models would materially improve those results as well. Clearly access to electricity is a prerequisite for internet access and thus is more important! This leads us to several thoughts:
  - a) This study would require a deeper analysis of the collinear relationship between electricity access and internet access with proper adjustments made

- b) Our models might better pinpoint the impact of internet access if we broke the data set into separate clusters and models based on electricity access as a control variable
- c) As described in our Project Proposal, the Rural Electrification Act and its impact on rapid development in the US was our inspiration for this study. We had assumed that investment in internet infrastructure was the modern day equivalent to electricity. However, these model results suggest that assuming access to electricity is a given may be quite faulty. This may lead to a prescriptive recommendation that digital companies and aid organizations might want to take a multi-pronged approach to addressing the “Next Billion Users” opportunity by investing in electricity infrastructure alongside, and as a precursor to, internet infrastructure

## **Next Steps**

Most of our “next steps” are outlined in the table and narrative above. Additional comments:

- To enhance our analysis and potentially improve predictive performance, we may expand our model selection to include deep learning feed-forward neural networks (FFNNs). This decision is based on these algorithms’ proven ability to capture complex, non-linear relationships within the data that simpler models might miss, including potentially addressing the overestimation / underestimation pattern. The primary FFNN architecture we would implement is the Multi-Layer Perceptron (MLP) which takes advantage of multiple options for activation and optimization functions (Jaiswal, 2025) . This will serve as our baseline for deep learning, and variations to this architecture through increasing the number of hidden layers or neurons will be used to assess the impact on performance.

Our validation strategy for these deep learning models will involve a robust k-fold cross-validation approach. This will provide a more reliable and less biased estimate of the model's performance on unseen data, which will help avoid the consequences of overfitting to a

single training-validation split. A vital part of the next steps in our plan is the comprehensive hyperparameter tuning for these networks by systematically evaluating a range of optimizers, including Adam, RMSprop, and SGD to see which of the options successfully navigates the issues incurred by our training and test data (Geeks for Geeks, 2025a). For activation functions, we'll experiment with ReLU, Sigmoid, and Tanh to compare performance against each other and Leaky ReLU which can help mitigate the "dying ReLU" problem. In addition, we will tune other key hyperparameters, including the learning rate, the number of hidden layers and neurons, the batch size, and the dropout rate to regularize the model and prevent it from becoming too specialized to the training data (Geeks for Geeks, 2025d). This focused exploration of feed-forward networks and their critical components represents a significant and intentional change from our initial analysis plan, which will allow us to fully leverage the power of deep learning for our predictive task. This process is expected to yield more accurate and generalized models compared to the initial predictive models to provide a more detailed solution to our analysis.

- We will explore alternative two-variable models that may improve performance
- We will build additional models using lagged internet predictors and (potentially) “change” variables to see if we can find stronger predictive correlation between investment in internet access and
- We may build multi-variate statistical predictive models
- We will examine the collinearity between electric access and internet access, and take appropriate actions to take advantage of what we’ve learned about the importance of electricity access as a precedent to internet access as twin drivers of future HDI improvement
- We will research whether it’s possible to use Principal Components in our modeling and still isolate the impact of a single predictor (internet access)

## **Appendix A**

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## Appendix B

### *Data Dictionary*

| Short Name              | Category / "Topic" | Long Name | Type       | Source            |
|-------------------------|--------------------|-----------|------------|-------------------|
| HDI_Category            | N/A                | N/A       | Response   | UNDP              |
| Lag1_InternetUsersPct   | N/A                | N/A       | Predictor  | World Bank (Data) |
| Lag2_InternetUsersPct   | N/A                | N/A       | Predictor  | World Bank (Data) |
| YearlyChgInternet       | N/A                | N/A       | Predictor  | World Bank (Data) |
| Lag1_YearlyChgInternet  | N/A                | N/A       | Predictor  | World Bank (Data) |
| Lag2_YearlyChgInternet  | N/A                | N/A       | Predictor  | World Bank (Data) |
| GovernanceScore         | N/A                | N/A       | Predictor  | World Bank (Data) |
| HealthSpend             | N/A                | N/A       | Predictor  | World Bank (Data) |
| seriesCode <sup>1</sup> | N/A                | N/A       | Predictor  | ITU - Mobile      |
| entityIso               | N/A                | N/A       | Predictor  | ITU - Mobile      |
| dataValue               | N/A                | N/A       | Predictor  | ITU - Mobile      |
| dataYear                | N/A                | N/A       | Predictor  | ITU - Mobile      |
| seriesCode <sup>2</sup> | N/A                | N/A       | Predictor  | ITU - Fixed       |
| entityIso               | N/A                | N/A       | Predictor  | ITU - Fixed       |
| dataValue               | N/A                | N/A       | Predictor  | ITU - Fixed       |
| dataYear                | N/A                | N/A       | Predictor  | ITU - Fixed       |
| Country                 | N/A                | N/A       | Response   | UNDP              |
| Year <sup>3</sup>       | N/A                | N/A       | Response   | UNDP              |
| HDI Index               | N/A                | N/A       | Response   | UNDP              |
| Gini Coefficient        | N/A                | N/A       | Response   | UNDP              |
| Country Name            | N/A                | N/A       | Identifier | World Bank        |

1  
2<sup>nd</sup>  
3

| Short Name        | Category / "Topic"                   | Long Name                                                                                                       | Type                | Source               |
|-------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------|----------------------|
|                   |                                      |                                                                                                                 |                     | (Data)               |
| Country Code      | N/A                                  | N/A                                                                                                             | Identifier          | World Bank (Data)    |
| Income Level      | N/A                                  | N/A                                                                                                             | Predictor / Control | World Bank (Country) |
| Year <sup>5</sup> | N/A                                  | N/A                                                                                                             | Identifier          | World Bank (Data)    |
| SE.SEC.CUAT.PO.ZS | Education: Outcomes                  | Educational attainment, at least completed post-secondary, population 25+, total (%) (cumulative)               | Response            | World Bank (Data)    |
| SE.SEC.CUAT.UP.ZS | Education: Outcomes                  | Educational attainment, at least completed upper secondary, population 25+, total (%) (cumulative)              | Response            | World Bank (Data)    |
| SE.TER.CUAT.BA.ZS | Education: Outcomes                  | Educational attainment, at least Bachelor's or equivalent, population 25+, total (%) (cumulative)               | Response            | World Bank (Data)    |
| SE.TER.CUAT.DO.ZS | Education: Outcomes                  | Educational attainment, Doctoral or equivalent, population 25+, total (%) (cumulative)                          | Response            | World Bank (Data)    |
| SE.TER.CUAT.MS.ZS | Education: Outcomes                  | Educational attainment, at least Master's or equivalent, population 25+, total (%) (cumulative)                 | Response            | World Bank (Data)    |
| SE.TER.CUAT.ST.ZS | Education: Outcomes                  | Educational attainment, at least completed short-cycle tertiary, population 25+, total (%) (cumulative)         | Response            | World Bank (Data)    |
| AG.PRD.FOOD.XD    | Environment: Agricultural production | Food production index (2014-2016 = 100)                                                                         | Predictor           | World Bank (Data)    |
| EN.POP.DNST       | Environment: Density & urbanization  | Population density (people per sq. km of land area)                                                             | Predictor           | World Bank (Data)    |
| EN.POP.SLUM.UR.ZS | Environment: Density & urbanization  | Population living in slums (% of urban population)                                                              | Predictor           | World Bank (Data)    |
| SP.RUR.TOTL.ZG    | Environment: Density & urbanization  | Rural population growth (annual %)                                                                              | Predictor           | World Bank (Data)    |
| EG.ELC.ACCS.ZS    | Environment: Energy production & use | Access to electricity (% of population)                                                                         | Predictor           | World Bank (Data)    |
| ER.H2O.FWST.ZS    | Environment: Freshwater              | Level of water stress: freshwater withdrawal as a proportion of available freshwater resources                  | Predictor           | World Bank (Data)    |
| FX.OWN.TOTL.ZS    | Financial Sector: Access             | Account ownership at a financial institution or with a mobile-money-service provider (% of population ages 15+) | Predictor           | World Bank (Data)    |
| SN.ITK.MSFI.ZS    | Health                               | Prevalence of moderate or severe food insecurity in the population (%)                                          | Predictor           | World Bank (Data)    |

| Short Name        | Category / “Topic”                   | Long Name                                                                                                 | Type      | Source            |
|-------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------|-------------------|
| SH.XPD.CHEX.PC.CD | Health: Health systems               | Current health expenditure per capita (current US\$)                                                      | Predictor | World Bank (Data) |
| SH.XPD.GHED.PC.CD | Health: Health systems               | Domestic general government health expenditure per capita (current US\$)                                  | Predictor | World Bank (Data) |
| SH.STA.WASH.P5    | Health: Mortality                    | Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (per 100,000 population) | Predictor | World Bank (Data) |
| SP.DYN.LE00.IN    | Health: Mortality                    | Life expectancy at birth, total (years)                                                                   | Response  | World Bank (Data) |
| SH.UHC.SRVS.CV.XD | Health: Universal Health Coverage    | UHC service coverage index                                                                                | Predictor | World Bank (Data) |
| IT.NET.BBND.P2    | Infrastructure: Communications       | Fixed broadband subscriptions (per 100 people)                                                            | Predictor | World Bank (Data) |
| IT.NET.USER.ZS    | Infrastructure: Communications       | Individuals using the Internet (% of population)                                                          | Predictor | World Bank (Data) |
| GB.XPD.RSDV.GD.ZS | Infrastructure: Technology           | Research and development expenditure (% of GDP)                                                           | Predictor | World Bank (Data) |
| IS.SHP.GCNW.XQ    | Infrastructure: Transportation       | Liner shipping connectivity index (maximum value in 2004 = 100)                                           | Predictor | World Bank (Data) |
| SI.POV.GINI       | Poverty: Income distribution         | Gini index                                                                                                | Response  | World Bank (Data) |
| VC.IHR.PSRC.P5    | Public Sector: Conflict & fragility  | Intentional homicides (per 100,000 people)                                                                | Predictor | World Bank (Data) |
| CC.EST            | Public Sector: Policy & institutions | Control of Corruption: Estimate                                                                           | Predictor | World Bank (Data) |
| GE.EST            | Public Sector: Policy & institutions | Government Effectiveness: Estimate                                                                        | Predictor | World Bank (Data) |
| PV.EST            | Public Sector: Policy & institutions | Political Stability and Absence of Violence/Terrorism: Estimate                                           | Predictor | World Bank (Data) |
| RL.EST            | Public Sector: Policy & institutions | Rule of Law: Estimate                                                                                     | Predictor | World Bank (Data) |
| VA.EST            | Public Sector: Policy & institutions | Voice and Accountability: Estimate                                                                        | Predictor | World Bank (Data) |
| SM.POP.TOTL.ZS    | Social Protection & Labor: Migration | International migrant stock (% of population)                                                             | Predictor | World Bank (Data) |

## **Appendix C - Code Run and Plots**

*Provided Under Separate Cover*